# Minwoo Park

mwpark@engineering.upenn.edu
linkedin.com/in/minwoo-park1 | github.com/minpark830 | minpark830.github.io
U.S. citizen, eligible for security clearance | Open to relocation

## Summary

Electrical engineer (M.S., Penn, Dec 2026) in embedded software, hardware, controls (GNC), and robotics: C++/Python servo/motor control at Varian, embedded C firmware on FreeRTOS for a three-drone swarm, a live EE interview-practice web app, and PCB design, compliance testing, and test automation at Stryker. Seeking full-time embedded, electrical, controls, and software roles.

## Education

**University of Pennsylvania** Expected Dec 2026
Master of Science in Electrical Engineering, Concentration in AI; GPA: 3.62/4.00

**California Polytechnic State University, San Luis Obispo** Sep 2021 – Jun 2025
Bachelor of Science in Electrical Engineering, Minor in Computer Science; GPA: 3.96/4.00
Dean's List (Fall 2021 – Spring 2025); President's Honor List (2021 – 2025)

## Skills

**Programming:** C, C++, Python (NumPy, PyTorch, OpenCV, matplotlib), MATLAB/Simulink, Java, JavaScript, HTML/CSS, SQL, Bash scripting, Verilog/SystemVerilog, VHDL, RISC-V assembly, LabVIEW
**Embedded systems:** real-time embedded C/C++ firmware, RTOS (FreeRTOS), microcontrollers (STM32, Arm Cortex-M), interrupts, I2C, SPI, UART, Ethernet, TCP/IP, hardware-in-the-loop (HIL/SIL) testing, firmware debugging, hardware/software integration, Arduino, Raspberry Pi
**Guidance, navigation, and control (GNC):** control theory, linear systems, PID, feedforward, state-space control, servo/motor control, motion control, disturbance observers, trajectory generation, path planning, Kalman filter (EKF), sensor fusion, state estimation, convex optimization, PLC programming
**Robotics and ML:** ROS and ROS 2, kinematics, computer vision, UAVs (quadrotors), multi-agent and distributed systems, particle-filter SLAM, machine learning, reinforcement learning (PPO)
**Software engineering:** data structures and algorithms, object-oriented programming, Git, GitHub Actions (CI/CD), Docker, Linux, Make, unit testing, Agile, REST APIs, FastAPI, Microsoft Azure, AWS
**Hardware design:** Altium Designer (schematic capture, PCB layout, footprints, BOM), Eagle, LTspice/SPICE, analog and mixed-signal circuits, power electronics, digital signal processing (DSP), FPGA digital design (AMD Xilinx Vivado), soldering, Fusion 360 (CAD)
**Lab and test:** test equipment (oscilloscope, logic analyzer, multimeter, spectrum analyzer, network analyzer), hardware debugging, test automation, EMC and power compliance testing (IEC 60601)
**Certification:** FCC Amateur Radio Technician License | **Languages:** English, Korean, Spanish (limited)

## Work Experience

**Varian (Siemens Healthineers)** — Palo Alto, CA May 2026 – Present
*Embedded Software Engineer / Control Systems Intern*

- Built and unit-tested servo/motor control, optimization, and path and treatment-planning algorithms in C++ and Python for radiotherapy machines, improving servo performance and motion accuracy.
- Increased trajectory generation speed by 30% on average by optimizing the planning algorithms.

**Kumar Lab, University of Pennsylvania** — Philadelphia, PA Aug 2025 – Present
*Research Assistant (aerial robotics; avg. 10 hrs/week)*

- Developing a research paper on multi-agent payload control, including literature synthesis, problem formulation, and analysis of trajectory generation and controller performance.
- Built a Dockerized ROS 2 framework for reproducible Crazyflie trajectory and controller experiments.

**CPSquare (Cal Poly Cyber-Physical Systems Lab)** — San Luis Obispo, CA Sep 2024 – Jun 2025
*Research Assistant (drone swarms; avg. 10 hrs/week)*

- Co-developed leader–follower formation control (triangle and line formations) for a distributed 3-drone Crazyflie swarm, flight-tested in the lab.
- Wrote the leader drone's embedded C firmware apps as FreeRTOS tasks, including a peer-to-peer radio version (DTR token ring) that takes the ground station out of the formation loop.
- Built a Python FastAPI ground server with REST and WebSocket APIs that relays commands and positions between a web page and each drone's radio link (one Crazyradio 2.0 per drone).
- Co-authored *Framework for Multi-Agent Coordination and Distributed Localization in Micro-UAVs*.

**Cal Poly Department of Electrical Engineering** — San Luis Obispo, CA Jan 2025 – Jun 2025
*Instructional Teaching Assistant (avg. 15 hrs/week)*

- Graded and gave feedback on EE 302 (Classical Controls) and EE 342 (Classical Control Systems Lab).

**Stryker (Endoscopy)** — San Jose, CA Jun 2024 – Sep 2024
*Advanced Operations Electrical Engineering Intern*

- Built LabVIEW automation for the 1788 camera head line, cutting manufacturing time by up to 20%.
- Adapted Precision Light Source calibration software to the L11 line, saving 5% of floor space.
- Revised the 1788 quality-check software, improving reliability and cutting debugging time by 2+ weeks.

**Cal Poly Department of Electrical Engineering** — San Luis Obispo, CA Oct 2023 – Jun 2024
*Instructional Teaching Assistant (avg. 10 hrs/week)*

- Taught soldering, PCB design, and microcontroller fundamentals in EE 143 (Circuit Analysis Lab).

**Stryker (Interventional Spine)** — Portage, MI Jun 2023 – Sep 2023
*R&D Electrical Engineering Intern*

- Designed the schematic and PCB layout of an Arduino-integrated test board in Altium Designer, reducing MultiGen 2 splitter box testing time by 75%.
- Built the board's schematic symbols and footprints by hand, chose parts to power and cost requirements, and generated the bill of materials (BOM) for purchasing.
- Tested and verified EMC and power compliance of RF ablation console PCBAs to IEC 60601.

## Projects

**Ohm: EE Interview Practice Web App** — minpark830.github.io/ohm Jan 2026 – Present

- Built and launched a web app for electrical engineering interview practice, with a 75-problem path across circuit analysis, RC circuits, BJTs, MOSFETs/CMOS, digital logic, and HDL.
- Wrote its JavaScript, HTML, and CSS front end: interactive SVG circuit simulations, instant answer checks with hints and solutions from a JSON API, spaced-repetition review, and timed mock interviews.

**FotoMagic Full-Stack Web App (Software Engineering course)** Apr 2025 – Jun 2025

- Wrote the GitHub Actions CI/CD pipelines (lint, build, deploy to Azure), Cypress end-to-end tests, and ESLint/Prettier setup for a 5-person team's React and Flask web app; co-developed its React UI.

**Vision-Controlled Robot Manipulator (OpenManipulator-X)** Sep 2024 – Dec 2024

- Implemented forward/inverse kinematics and computer vision for autonomous pick-and-place sorting.

**Ultrasonic Underwater Sensor Network** Sep 2024 – Dec 2024

- Built an end-to-end underwater acoustic link carrying DHT11 sensor data through ultrasonic hardware and analog signal processing (AGC, PLL tone decoding) to a Raspberry Pi web server.

**Custom IR RC Car (EE 329 final project)** Apr 2024 – Jun 2024

- Built an IR-controlled RC car on an STM32 microcontroller, using I2C and SPI drivers and interrupts to integrate motor drivers, IR and ultrasonic sensors, an accelerometer, and an LCD.

**RISC-V 32-bit MCU** Jan 2024 – Mar 2024

- Implemented a pipelined RISC-V CPU with cache and peripheral I/O in SystemVerilog on an FPGA.

**4-Bit DAC PCBA (Digital-to-Analog Converter)** Apr 2022 – Jun 2022

- Designed a binary-weighted resistor DAC and integrated it with an Arduino speaker system.